Syllabus 9758 Statistics Example Questions

Discrete Random Variables

Expectation of a new variable

Example 1

[SAJC]

The number of muffins, X, bought by a randomly chosen customer at a muffin shop has probability distribution as shown in the table below:

x	1	2	3	4	5	6	>6
$P(X=x)$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	a	$\frac{2}{12}$	$\frac{1}{12}$	0

(i) Find the value of a.

(ii) State the expected value of X.

(iii) Evaluate the variance of X.

Initially, the shopkeeper charges \$2.50 for each muffin, which cost 50 cents to produce.

(iv) Using the answer to part (ii), or otherwise, find the expected profit made per customer.

The shopkeeper decides to promote the sale of muffins by giving a free muffin to all customers who buy 4 or more muffins. As a result, the probability distribution of the number of muffins, Y, taken away by a customer is given in the following table.

y	1	2	3	4	5	6	>6
$P(Y=y)$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	0	$\frac{20}{36}$	$\frac{6}{36}$	0

(v) Find the expected profit made per customer.

(vi) If the number of customers increases by 8%, is the promotion more profitable for the shopkeeper? Give a reason for your answer.

[Ans: (i) $\frac{1}{4}$ (ii) 3.5 (iii) $1\frac{11}{12}$ (v) \$6.86]